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LITTORAL PHYSICS TEACHERS' ASSOCIATION (LIPTA)

REGIONAL MOCK

CAMEROON GENERAL CERTIFICATE OF EDUCATION

0780 PHYSICS 1

MARCH 2025 ADVANCED LEVEL

Centre Number	
Centre Name	
Candidate Identification No	
Candidate's Name	

Mobile phones are NOT allowed in the examination room.

MULTIPLE CHOICE QUESTION PAPER

One and Half hours (1 ½ hours)

INSTRUCTIONS TO CANDIDATES

Read the following instructions carefully before you start answering the questions in this paper. Make sure you have a soft HB pencil and an eraser for this examination.

1. USE A SOFT HB PENCIL THROUGH OUT THE EXAMINATION.
2. DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

Before the examination begins

3. Check that this question booklet is headed '**Advanced Level – 780 PHYSICS 1**'.
4. Fill in the information required in the spaces above.
5. Fill in the information required in the spaces provided in the answer sheet using your HB pencil: **Candidate Name, Exam Session, Subject Code, Centre Number and name** and **Candidate Number**. Take care that you do not crease or fold the answer sheet or make any marks on it other than those asked for in these instructions.

How to answer the questions in this examination.

6. Answer **ALL** the **50 questions** in this examination. All questions carry the same mark.
7. Each question has FOUR suggested answers; **A, B, C** and **D**. Decide which answer is correct. Find the question number on the answer sheet and draw a horizontal line across the letter to join the square brackets for the answer you have chosen. For example, if **C** is the selected answer, mark **C** as shown below.

[A] [B] [C] [D]

8. Mark only one answer for each question. If you mark more than one answer, you will score zero for that question. If you change your mind about an answer, erase the first mark carefully, and then mark your new answer.
9. Avoid spending too much time on any question. If you find a question difficult, move on to the next question. You can comeback to this question later.
10. Do all rough work in this booklet, using, where necessary, the blank spaces in the question booklet.
11. **At the end of the examination, the invigilator shall collect the answer sheet first and then the question booklet. DO NOT ATTEMPT TO LEAVE THE EXAMINATION HALL WITH IT.**
12. The use of non-programmable calculators and GCE Board formulae booklets is allowed.

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SECTION I (Thirty-five questions)
Questions 1 – 35
Directions:

Each of the questions or incomplete statements in this section is followed by four suggested answers. Select the best answer for each question.

1) A physically equation;
   A) has the correct unitless constants and correct terms having quantities with the same units.
   B) possesses units of quantities, unitless constants and correct terms.
   C) possesses constants with units, correct terms and units.
   D) has terms with the same units, correct unitless constants and correct terms.

2) The following physical quantities have the same units.
   A) Impulse and rate of change of momentum
   B) Angular frequency and angular velocity
   C) Pressure and strain
   D) Potential gradient and potential difference

3) Figure 1 shows three coplanar forces acting on a body.

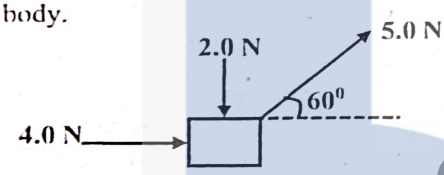


Figure 1

The resultant force on the body is;
   A) 6.9 N at an angle of 19.7^0 above the x-axis.
   B) 6.9 N at an angle of 19.7^0 below the x-axis.
   C) 9.1 N at an angle of 44.2^0 above the x-axis.
   D) 9.1 N at an angle of 44.2^0 below the x-axis.

4) Figure 2 shows a constant force F acting on a body of mass m. The body is placed on a smooth inclined plane at an angle, θ, with the horizontal.

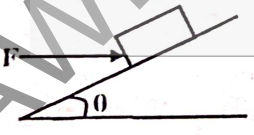


Figure 2

The resultant force acting along the plane on the object is;
   A) Fsinθ – mgsinθ
   B) Fsinθ – mgcosθ
   C) Fcosθ – mgcosθ
   D) Fcosθ – mgsinθ

5) A long jumper, jumps at an angle of 20^0 to the horizontal with a take-off speed of 11.0 m s^{-1}. The distance of his jump is;
   A) 3.96 m
   B) 4.22 m
   C) 7.93 m
   D) 12.33 m

6) A particle P moves with constant velocity and collides head-on with a stationary particle Q. After the collision, both particles move with the same velocity. From the collision, it is true that;
   A) the increase in momentum of Q is equal to the decrease in momentum of P.
   B) the increase in kinetic energy of Q is equal to the decrease in kinetic energy of P.
   C) the total kinetic energy of P and Q is greater than the total kinetic energy before the collision.
   D) the force acted by Q on P is less than the force acted by P on Q during the collision.

7) Figure 3 shows how the power (P) of a motorcycle varies with its velocity (v).

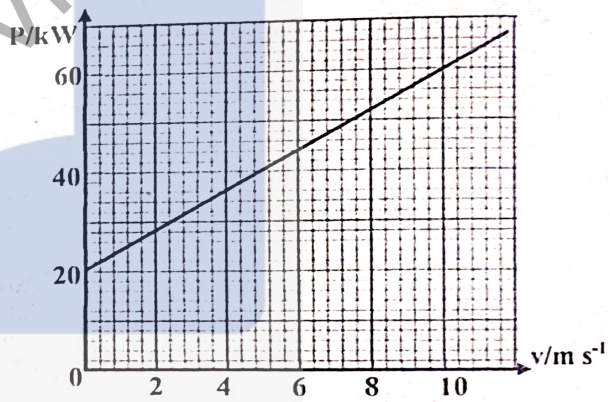


Figure 3

If the mass of the motorcycle is 425 kg and that of the rider is 75 kg, then the acceleration of the motorcycle is;
   A) 1.25 m s^{-2}
   B) 4.00 m s^{-2}
   C) 8.00 m s^{-2}
   D) 12.5 m s^{-2}

8) When a system is critically damped;
   A) no energy is lost from the system
   B) the system takes an infinite time to return to equilibrium

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C) only a few oscillations are possible
D) the system does not oscillate but returns easily to the equilibrium position

9) Figure 4 shows a particle moving with a constant speed of 10 m s^{-1} in a horizontal circular path about a point O.

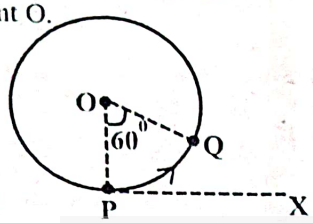


Figure 4

The change in velocity of the particle in its motion from point P to point Q is;
A) 5.0 m s^{-1} in the direction of PX
B) 7.3 m s^{-1} at an angle of 30^0 to the direction PX
C) 8.7 m s^{-1} at an angle of 90^0 to the direction PX
D) 10.0 m s^{-1} at an angle of 120^0 to the direction PX

10) Figure 5 shows the variation of pressure (P) of an ideal gas with density (ρ) at temperatures T and 300 K.

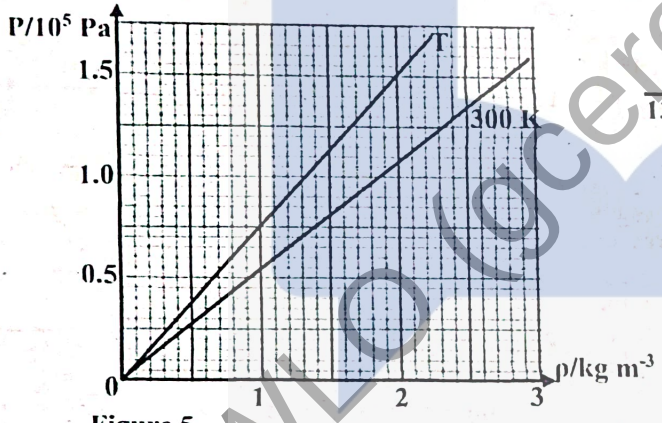


Figure 5

The value of the temperature T is;
A) 214 K
B) 420 K
C) 355 K
D) 254 K

11) Given that the molar heat capacity at constant volume for a gas is kR where k is a constant and R is the universal gas constant. The ratio of the molar heat capacity at constant pressure to the molar heat capacity at constant volume will be;
A) $\frac{k}{k+1}$
B) $\frac{k+1}{k}$
C) $\frac{k}{k-1}$
D) $\frac{k-1}{k}$

12) Figure 6 shows the variation of pressure P with volume V for a given mass of a gas.

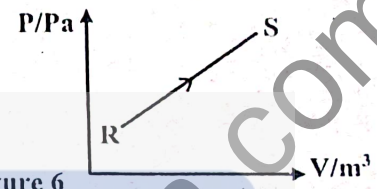
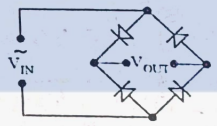


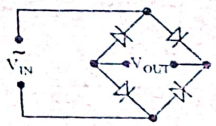
Figure 6

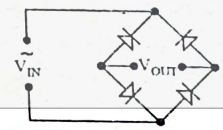
When the gas changes state from R to S then;

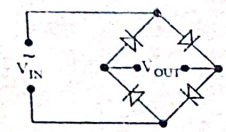
	Thermal energy transfer	Internal energy	Work done
A)	Supplied to gas	Increase	By gas
B)	Supplied to gas	Decrease	On gas
C)	Released by gas	Increase	On gas
D)	Released by gas	Increase	By gas

13) Which one of the following circuits best shows how the diodes should be arranged for a bridge rectifier to operate correctly?

A.


B.


C.


D.


14) A container has a layer of kerosene above a layer of acetone. Light is incident through the acetone of refraction index 1.36 and is completely polarized at the acetone-kerosene boundary at an angle of 45.63^0. The refractive index of kerosene is;
A) 1.33
B) 1.20
C) 1.39
D) 1.54

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15) Figure 7 shows the stress-strain graph for two types of steel P and Q.

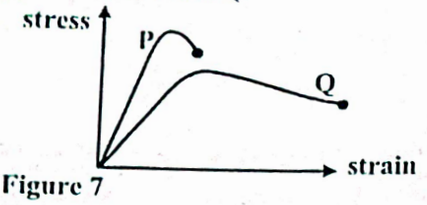


Figure 7

The best statement that describes the relative properties of P and Q is;

A) P has greater stiffness and lower toughness than Q.
B) P has lower stiffness and lower toughness than Q.
C) P has lower stiffness and greater toughness than Q.
D) P has greater stiffness and greater toughness than Q.

16) The specific heat capacity of a gas at constant pressure differs from the specific heat capacity of the gas at constant volume because;

A) additional heat is needed for the expansion of the gas at constant pressure
B) additional heat is needed to increase the number of degrees of freedom of the gas at constant volume.
C) More work has to be done to overcome the stronger attractive forces between molecules at constant pressure.
D) the internal energy of the gas at constant pressure is higher than the internal energy of the gas at constant volume.

17) Figure 8 shows two stars X and Y of masses m_1 and m_2 respectively and separated by a distance r.

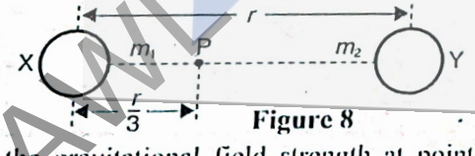


Figure 8

If the gravitational field strength at point P at distance r/3 from X is zero, the ratio of m_1 to m_2 is

A) 1 : 4
B) 1 : 2
C) 2 : 1
D) 4 : 1

18) Figure 9 shows a sound source S moving away from observer P towards observer Q.

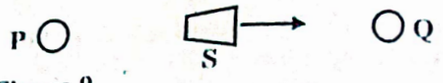


Figure 9

What is the effect of the motion on the wavelength of sound received by P and by Q?

	P	Q
A)	Increases	Decreases
B)	Increases	No change
C)	Decreases	Increases
D)	No change	Decreases

19) The graph in figure 10, shows the variation of the interatomic force F with the separation r between two atoms.

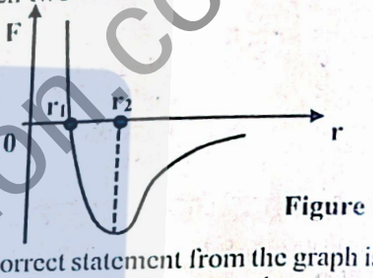


Figure 10

The incorrect statement from the graph is that;

A) Hooke's law is true around r_1.
B) the equilibrium separation is r_2.
C) the force is an attractive force when $r > r_1$.
D) the force is an attractive force when $r > r_2$.

20) A cylinder of volume 0.09 m^3 contains an ideal gas at 100 °C and 8×10^4 Pa. If the mass of one mole of the gas is 0.034 kg, the r.m.s speed of the gas molecules is;

A) 2.71×10^2 m s^{-1}
B) 7.33×10^4 m s^{-1}
C) 5.23×10^2 m s^{-1}
D) 2.73×10^5 m s^{-1}

21) In an experiment to measure the thermal conductivity of a good conductor, a long test sample is used so that

A) the heat loss can be neglected
B) the steady state is easily achieved
C) the temperature difference between the two ends of the sample is sufficiently large
D) the heat flow is parallel throughout the sample

22) Figure 11 shows an uninsulated uniform metal rod.

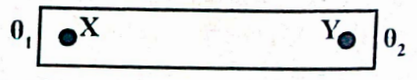


Figure 11

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Heat flows through the rod in a steady state, and the temperatures at the ends of the rod are θ_1 and θ_2, where $\theta_1 > \theta_2$. The true statement of the two positions X and Y on the rod is;

A) the rate of heat loss to the surroundings at X is greater than that at Y.
B) negligible heat is lost to the surroundings at the area between X and Y.
C) the rates of heat flow at X and Y are the same.
D) X and Y have the same temperature gradients.

23) A lamp lights up immediately when the switch in a circuit is closed, although the drift velocity of electrons is small. This is because

A) the current flowing in the circuit is large
B) the potential difference supplied in the circuit is large
C) all the free electrons drift simultaneously in the circuit
D) the random velocity of the electrons is large

24) A diffraction grating with 2400 lines per centimetre is 0.625 m from a detector and used to produce a diffraction pattern. The first-order fringe is 8.94 cm from the central fringe.

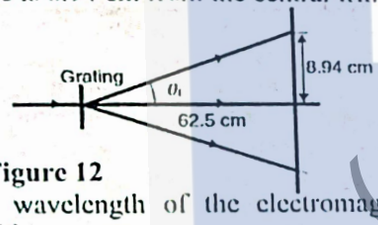


Figure 12

The wavelength of the electromagnetic wave used is;

A) 295 nm
B) 590 nm
C) 1180 nm
D) 1190 nm

25) Which statement is not true of the back e.m.f of an electric motor?

A) the back e.m.f helps increase the magnitude of the current flowing through the coil of the motor.
B) the back e.m.f opposes the applied voltage.
C) the magnitude of the back e.m.f increases if the speed of rotation of the coil of the motor is increased.
D) the back e.m.f is caused by electromagnetic induction.

26) If the masses of the proton, neutron, electron and $^{12}_{6}C$ atom are m_p, m_n, m_e and m_c respectively, the mass defect Δm of the nucleus $^{12}_{6}C$ is given by;

A) $\Delta m = 6m_p + 6m_n - m_c$
B) $\Delta m = 6m_p + 6m_n - 6m_e - m_c$
C) $\Delta m = 6m_p + 6m_n + 6m_e - m_c$
D) $\Delta m = 6m_p + 6m_n + 6m_e - 6m_c$

27) A sample consists of 2 g of a radioactive element. The molar mass of the element is 67 g. If the half-life of the element is 78 hours, the activity of the sample after 48 hours is

A) $2.7 \times 10^{16}\ s^{-1}$
B) $2.9 \times 10^{16}\ s^{-1}$
C) $4.4 \times 10^{16}\ s^{-1}$
D) $1.0 \times 10^{20}\ s^{-1}$

28) Figure 13 shows the energy levels of the hydrogen atom.

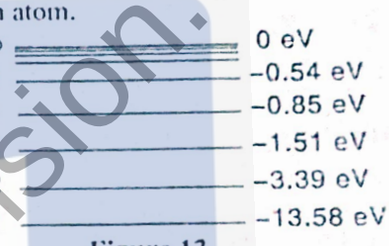


Figure 13

The transition that produces a radiation of wavelength 436 nm is;

A) n = 4 to n = 1
B) n = 5 to n = 2
C) n = 4 to n = 2
D) n = 5 to n = 1

29) Figure 14 shows the path of an α-particle as it approaches a massive nucleus at N.

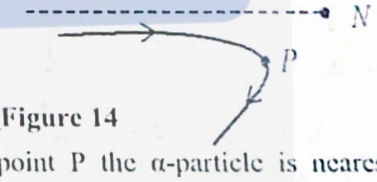


Figure 14

At point P the α-particle is nearest to the nucleus. Then;

A) the electric potential energy of the α-particle is minimum and the total energy is maximum.
B) the angular momentum of the α-particle about N is minimum.
C) If the initial kinetic energy of the α-particle was greater, the distance between P and N would be larger.
D) If the atomic number of the nucleus was greater, the distance between P and N would be larger.

30) The correct statement about a stationary wave is that;

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A) between two nodes the amplitude of the wave is constant.
B) between two nodes all parts of the wave vibrate in phase.
C) the two waves producing the stationary wave must always be 180° out of phase.
D) the separation of the nodes for the second harmonic is double the separation of nodes for the first harmonic.

31) Figure 15 shows a transistor circuit with a base-emitter voltage of 0.6 V when the transistor conducts and a current amplification factor of the transistor is 100.

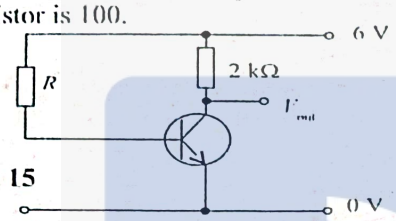


Figure 15

The value of R when the output voltage is 2 V, will be;
A) 270 kΩ
B) 150 kΩ
C) 300 kΩ
D) 540 kΩ

32) Figure 16 shows a circuit having cells of negligible internal resistance.

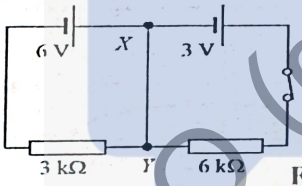


Figure 16

The potential difference between the points X and Y and current in the wire connecting the two points is;

	Potential difference	Current
A)	0 V	1.5 mA
B)	0 V	0.0 mA
C)	3 V	1.5 mA
D)	3 V	0.5 mA

33) A parallel plate capacitor is charged by connecting it to a battery and the energy stored in the capacitor is U_0. With the battery still connected, the space between the plates is filled with an insulator of dielectric constant 4.0. The energy stored in the capacitor will be;
A) (1/16) U_0
B) (¼) U_0
C) 4 U_0
D) 16 U_0

34) Figure 17 shows a thin lens made from a material of refractive index 1.48.

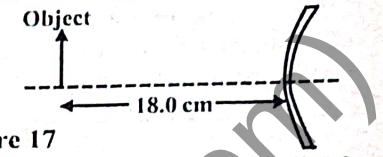


Figure 17

It has spherical surfaces of radii 5.0 cm and 3.5 cm. The nature and position of an image formed when an object is placed 18.0 cm to left of the lens will be;
A) A real image at 5.63 cm to the right of the lens.
B) A real image at 10.3 cm to the right of the lens.
C) A virtual image at 5.63 cm to the left of the lens.
D) A virtual image at 10.3 cm to the left of the lens.

35) A diffraction grating has 6000 lines per centimetre. When monochromatic light is incident normally on the grating, the first order diffracted beam makes an angle of 15^0 with the direction of the incident beam. The frequency of the incident light is;
A) 4.7×10^{13} Hz
B) 1.9×10^{14} Hz
C) 3.6×10^{14} Hz
D) 7.0×10^{14} Hz

SECTION II (Ten questions)

Questions : 36 - 45

Directions: For each group of questions below, ONE or TWO of the responses given is/are correct. Then choose.
A) If 1 and 2 are correct
B) If 2 and 3 are correct
C) If 1 only is correct
D) If 3 only is correct

Directions Summarized			
A	B	C	D
1, 2 correct	2 ,3 correct	1 only	3 only

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36) The table below gives the corrected count rate (in counts per minute) from three samples of radioactive isotopes at three different times.

Isotope	0 min	20 min	40 min
X	480	243	119
Y	135	32	9
Z	168	118	93

The above data show that;
1. X produces the most penetrating radiation.
2. Y has the largest decay constant.
3. Z has the longest half-life.

37) In figure 18, A and B are identical light bulbs and L is a pure iron cored inductor.

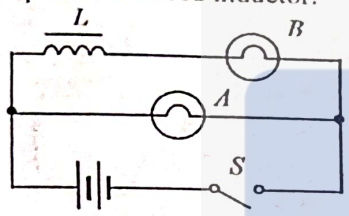


Figure 18

Which of the following statements is/are correct?
1. When switch S is closed, bulb A will light up first.
2. After switch S is closed for some time, A and B are equally bright.
3. When switch S is open, bulb A will go out first.

38) A simple motor rotating at a steady speed is suddenly jammed and comes to a stop. Which of the following statements is/are correct?
1. The rate of change of magnetic flux through the coil of the motor will be very large.
2. A large e.m.f will be induced in the coil of the motor.
3. A large current will flow through the coil of the motor.

39) The circuit in figure 19, shows a battery E of internal resistance 4 Ω connected in series to a variable resistor R.

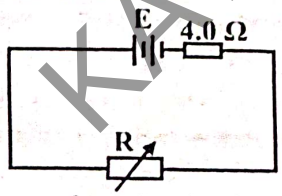


Figure 19

It is true from the circuit that;
1. If the resistance of R is very small, the terminal p.d of the battery is very small and the power dissipated in the battery is small.
2. If the resistance of R is very large, the current in the circuit is very small but most of the power supplied by the battery is dissipated by R.
3. If the resistance of R is 4 Ω, the power supplied to R by the battery is maximum

40) For a real gas;
1. the volume of the molecules cannot be neglected.
2. intermolecular forces may be neglected except during collisions.
3. collisions between molecules and the wall of a container are not perfectly elastic.

41) Figure 20 shows two small test charges +q and +2q brought from infinity to the positions shown.

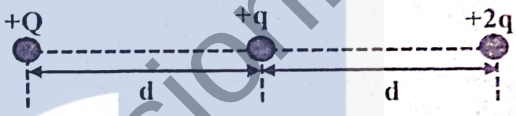


Figure 20

The two charges are collinear with another charge +Q and their mutual separation is d. It is true from the figure that;
1. the potential energy of the system would increase if d increases.
2. the work done in bringing the charges +q and +2q from infinity to their respective positions is the same.
3. charge +q is at a higher potential than charge +2q.

42) When a beam of white light passes through iodine vapour, the spectrum of the emergent light shows dark lines. Which of the following statement(s) is/are correct?
1. The iodine vapour absorbs all frequencies from the light except those which it emits.
2. The iodine vapour emits less energy than it absorbs.
3. The iodine vapour absorbs the same frequencies as it emits.

43) An air bubble is blown at the end of a capillary tube dipping into a liquid of surface tension T. The excess pressure needed to blow the bubble is;
1. directly proportional to T.
2. greatest at the beginning when one starts to blow.
3. least when the bubble is a hemisphere.

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44) In a simple astronomical telescope under normal adjustment;
1. the first image is formed at the focal plane of the objective.
2. the first image is real and inverted.
3. the focal length of the objective is almost the same but smaller than that of the eyepiece.

45) In an evacuated tube, a beam of electrons is accelerated through a p.d, V, and strike a graphite sheet X.

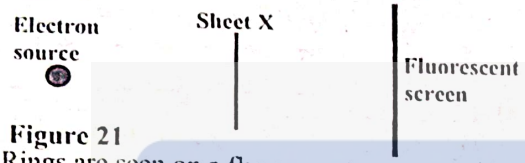


Figure 21

Rings are seen on a fluorescent screen positioned behind X. From this setup it is true that;
1. the sheet acts as a diffraction grating
2. the rings indicate the wave nature of electrons
3. when V is increased, the rings are wider apart.

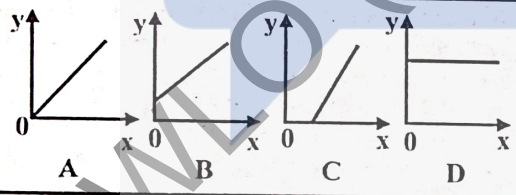
SECTION III (Five questions)

Questions : 46 – 50

Directions: *Each of the questions in this section has four sets of graphs A – D. Which of the graphs in each question best fits the relationship between x and y.*

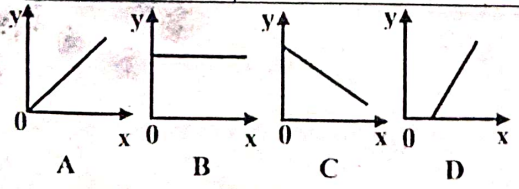
46)

y	x
Reciprocal of magnification of an object	Distance of the object from a convex lens.



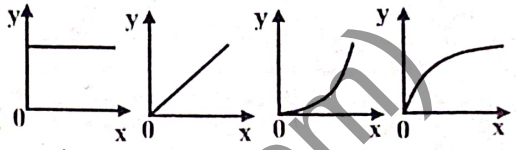
47)

y	x
Work function of a metal.	Frequency of light incident on the metal surface



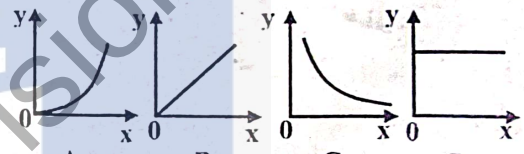
48)

y	x
Momentum of a car that accelerates uniformly from rest in a straight line.	Distance travelled by the car



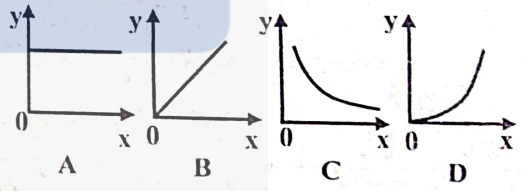
49)

y	x
Self-inductance of a solenoid.	Length of the solenoid.



50)

y	x
Force on an electron in the electric field between to oppositely charged large parallel plates.	Perpendicular distance of the electron from the positive plate.



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END

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